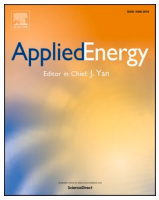
Applied Energy 385 (2025) 125569



Contents lists available at ScienceDirect

Applied Energy

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A novel multivariable prognostic approach for PEMFC degradation and remaining useful life prediction using random forest and temporal convolutional network

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HIGHLIGHTS

- A novel multivariable data-driven approach is proposed for PEMFC prognostics.
- The RF-RFE method is employed to automatically select optimal features.
- The baseline TCN is enhanced to develop an improved predictive model.
- PSO is utilized to automate hyperparameter configurations.
- By leveraging the optimal features selected, the developed predictive model achieves greater accuracy and demonstrates superior performance.

ARTICLE INFO

Keywords:
Proton exchange membrane fuel cell
Prognostics
Future degradation trend
Remaining useful life
Temporal convolutional network
Random forest

ABSTRACT

Data-driven methods are effective in predicting future degradation trends (FDT) and remaining useful life (RUL) of proton exchange membrane fuel cells (PEMFCs). However, the complex and dynamic degradation behaviour of PEMFCs, influenced by diverse operational variables, poses significant challenges to existing prognostic approaches. This paper proposes a novel multivariable prognostic approach, termed RF-TCN, which combines random forest (RF) with temporal convolutional networks (TCN) to address these challenges. The approach incorporates three key innovations: (1) A hybrid RF and recursive feature elimination (RFE) method is employed to automatically select features most relevant to fuel cell degradation, reducing manual intervention and enhancing input robustness. (2) An improved TCN-based model is developed to effectively capture temporal degradation patterns, enabling accurate FDT and RUL predictions. (3) Particle swarm optimization (PSO) is utilized for automatic hyperparameter configuration, further boosting predictive performance. Empirical validation on ageing durability datasets demonstrates that the RF-TCN approach achieves superior prediction accuracy with selected optimal features and outperforms baseline TCN, CNN, RNN, and existing methods in the literature. This work advances prognostic methodologies, contributing to extending fuel cell lifespan and optimizing control strategies.

1. Introduction

1.1. Background

Driven by the dual imperatives of global energy transition and environmental protection, proton exchange membrane fuel cells (PEMFCs) are increasingly recognized as a vital component of future energy systems because of their ability to convert energy efficiently and cleanly [1,2]. However, their reliability and durability remain insufficient for large-scale commercial deployment [3–5]. Prognostics and

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https://doi.org/10.1016/j.apenergy.2025.125569
Received 8 October 2024; Received in revised form 27 January 2025; Accepted 15 February 2025
Available online 19 February 2025
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Table 1
Existing data-driven models for PEMFC degradation prognostics.

Types	Models	Input feature	Feature selection	Optimizer	Year, Refs.
CNNs	Conditional CNN	Voltage	–	–	2024, [5]
	Attention-based CNN; Conditional CNN	Voltage & OVs	Manually	Adam & grid search	2022, [19]
	1D CNN	Voltage & OVs	Cat Boost & Manually	Adam	2024, [27]
RNNs	Attention-based LSTM/GRU	Voltage & OVs	Manually	Adam & grid search	2021, [18]
	GRU	Voltage & OVs	RF & Manually	Bayesian	2023, [28]
	Attention-based BiLSTM	Voltage & OVs	RF & Manually	DE algorithm	2020, [29]
	BiLSTM	Voltage	–	Bayesian	2023, [21]
	LSTM, GRU	Voltage & OVs	–	Tanh/Adam	2023, [22]
	TCN-LSTM	Voltage	–	Adam	2024, [16]
	ESN	Voltage & OVs	SNR & Manually	Manually	2020, [30]
	SI-ESN, DI-ESN	RPLR & OVs	Manually	Sensibility analysis	2021, [26]
	LSTM, GRU	RVLR	–	–	2024, [43]
	Transformer	Voltage	–	Adam	2024, [23]
	TLTNN	Voltage & OVs	Boruta & Manually	–	2023, [31]
	RCLMA	Voltage	–	Adam	2023, [24]
	TFT	Voltage & OVs	Covariate analysis	Adam	2023, [32]
Others	ANFIS	Voltage	–	PSO	2019, [15]
	GNNM	Voltage & OVs	Manually	PSO	2019, [33]
	DBN-ELM	Voltage	–	PSO	2020, [25]
	SVR-GRU-RELM	Voltage & OVs	MIC & AIC	IGOA	2022, [17]
	MP-SRCDKF	Voltage	–	–	2023, [20]

Note: "OVs" refers to operational variables, such as density, temperature, humidity, pressure and so on.

health management (PHM) is a field focused on estimating the health state of systems and providing useful information to users before failure occurs [6]. PEMFC prognostics aims to predict future degradation trends (FDT) and remaining useful life (RUL) by analyzing current and historical data [7,8]. FDT represents the projected trajectory of performance degradation over time [7], while RUL estimates the remaining operational duration before performance drops below a predefined threshold or fails to meet functional requirements [9]. Accurate predictions of FDT and RUL enable the early detection of degradation signs, allowing engineers to implement timely maintenance or adjust operational strategies to avoid sudden failures and improve the durability of PEMFCs [10–12]. Therefore, extensive research has been dedicated to improving FDT and RUL prediction accuracy, optimizing fuel cell management and extending their lifespan.

1.2. Literature review

PEMFC degradation prognostics are commonly addressed using model-based, data-driven, and fusion-based approaches [13,14]. Model-based methods rely on high-fidelity physical models [15], but their applicability is limited by the complexity and incomplete understanding of PEMFC degradation mechanisms [16]. Fusion-based methods, though fundamentally model-based, are challenging to model and require further theoretical development [17]. In contrast, data-driven methods have gained attention for their ability to predict PEMFC degradation directly from experimental data, eliminating the need for complex physical models [18]. Therefore, developing data-driven methods becomes a promising solution to addressing prediction challenges in fuel cell energy systems [9,19].

Currently, a variety of data-driven models have been applied to predict the FDT and RUL of PEMFCs. As summarized in Table 1, these models can be classified into two categories based on the number of input features: univariate and multivariable models. Univariate models [5,15,16,20–25], which primarily use stack voltage as the input, are computationally simple but fail to account for the influence of multiple operational variables. Additionally, the reliance on a single variable susceptible to noise diminishes prediction accuracy and reliability. By contrast, multivariable models [17–19,22,26–33] incorporate a range of operational parameters, such as current density, temperature, humidity, gas pressure, and fuel purity, showing more competitive and enabling more robust and accurate predictions, especially in complex, dynamic PEMFC systems [34,35]. However, despite their advancements, several critical challenges persist.

First, many studies rely on manually selected input features [19,27–31,33], which may result in inappropriate feature subsets and adversely affect prediction performance. This issue is critical, as the performance of data-driven methods largely depends on the selection of input features [36,37]. For instance, Zhang et al. [17] used the maximum information coefficient and Akaike information criterion (MIC-AIC) method to automate feature selection. However, their approach included performance output variables of individual cells, which limits its ability to fully exploit operational variables, and measuring these variables in practice is challenging [38]. Ensemble methods that integrate multiple single-feature selection algorithms [39], such as support vector machine-recursive feature elimination (SVM-RFE) [40] and random forest-recursive feature elimination (RF-RFE) [41], have demonstrated improved accuracy in areas like gene selection for cancer classification [40] and molecular biology dataset analysis [41]. Nevertheless, these methods have not yet been applied to multivariable predictive models for fuel cells. RF-RFE combines the strengths of RF in handling high-dimensional data with complex nonlinear relationships and providing fair feature importance assessments [42], alongside the iterative elimination capability of RFE. Leveraging these advantages, this study employs RF-RFE to identify features correlated with fuel cell degradation while excluding performance output variables of individual cells to fully utilize operational variables.

Second, most existing methods are primarily based on convolutional neural networks (CNNs) and recurrent neural networks (RNNs), which have inherent limitations [14]. CNNs are inefficient at capturing long temporal dependencies due to the limitations of convolution kernel size [6,11], while RNNs suffer from short-term memory issues, vanishing and exploding gradients, and high computational costs [14,23]. To track these shortcomings, various strategies have been explored, such as integrating attention mechanisms into CNN/RNN architectures [18,19,29] and creating ensemble learning-based stacking models [17]. However, attention mechanisms generally increase model complexity [18,23], while stacking models face challenges in reducing model redundancy [44]. Transformer-based models, despite their exceptional performance in natural language processing, computer vision, and speech processing [14], demand significantly larger datasets and greater computational resources, limiting their applicability in data-constrained scenarios [45]. In contrast, the temporal convolutional network (TCN) is specifically designed for time series modelling [46]. By utilizing dilated causal convolutions, TCN naturally captures long-term dependencies in time series data while maintaining lower computational complexity compared to transformer-based models [14,46]. Additionally, TCN offers a flexible architecture adaptable to diverse prediction tasks. Nevertheless, the baseline TCN [46] often requires modifications to

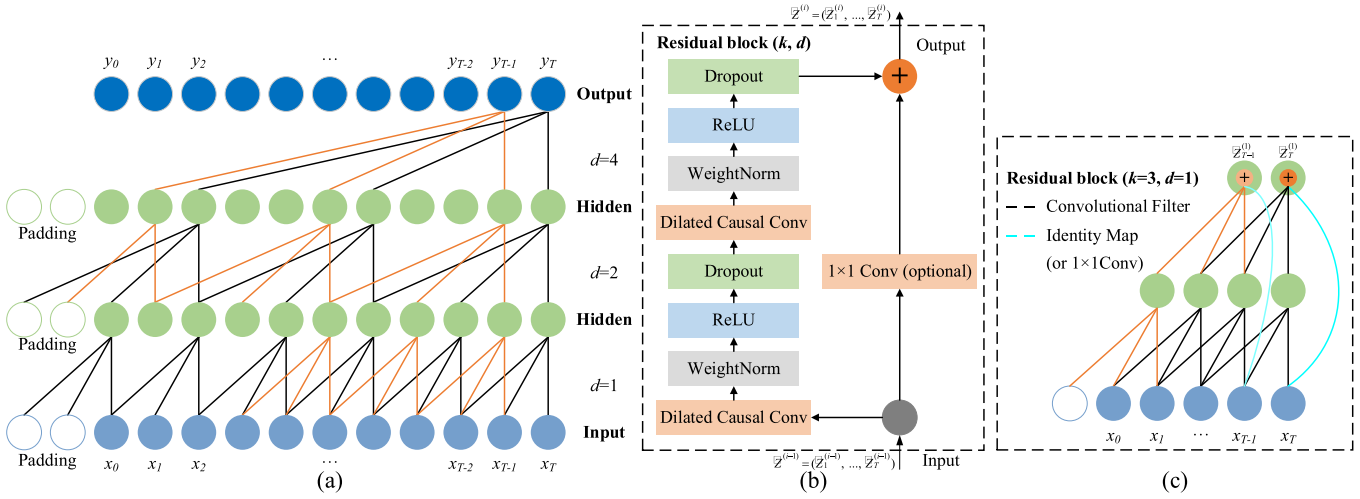


Fig. 1. The architecture of the TCN [46]: (a) dilated causal convolution; (b) residual block; (c) an example of a residual block.

adjust its receptive field size for specific prediction tasks. For instance, Wang et al. [47], Zhang et al. [48], and Fu et al. [49] employed repeated residual blocks to enhance feature extraction capabilities, but this approach significantly increases computational costs. In light of the above, this study employs the TCN for PEMFC degradation and RUL prediction, while introducing enhanced strategies to improve its performance.

Third, configuring hyperparameters is a vital aspect of optimizing data-driven models [25,50], as manually or empirically determined parameters often fail to deliver optimal performance. Common techniques include grid search [18,19], Bayesian optimization [21,28], and intelligent optimization algorithms [15,17,25,33]. Among these, intelligent optimization algorithms are particularly effective due to their capability to locate global optimal with fast convergence and high precision [51]. Particle swarm optimization (PSO) [52], recognized for its simplicity, efficient search capability, and effective use of memory, is well-suitable for optimizing complex models [15,33]. This study employs PSO to automate hyperparameter configuration, ensuring efficient exploration of parameter search spaces. Hyperparameter search ranges are derived from existing literature, with the optimization process conducted on validation datasets. The final model performance is assessed on independent test datasets.

1.3. Research gap and contributions

The prediction accuracy of existing multivariable methods for PEMFC systems is limited by several limitations: First, reliance on manual feature selection often results in suboptimal subsets, compromising predictive performance. Second, conventional CNNs and RNNs, commonly used in prognostic modelling, struggle to capture long-term dependencies and complex temporal patterns due to architectural constraints. Third, empirical hyperparameter tuning often yields suboptimal configurations, reducing model accuracy.

To address these challenges, this paper proposes a novel multivariable prognostic approach that combines RF and TCN, termed RF-TCN, to enhance the prediction accuracy of FDT and RUL for PEMFC systems. The approach introduces three core components: (1) The RF-RFE method is employed for the automated selection of features most relevant to degradation, minimizing manual intervention and improving input robustness. (2) An improved TCN architecture is developed to effectively capture complex temporal degradation patterns and long-term dependencies. (3) PSO is employed to automate hyperparameter tuning, ensuring optimal configurations and improved model performance.

The key innovations and contributions of this study are as follows:

(1) This study is the first to apply RF-RFE for PEMFC systems, enabling automated selection of optimal features and achieving higher accuracy compared to single-feature or all-feature configurations.
(2) Enhanced strategies are introduced into the baseline TCN, resulting in a predictive model for fuel cells that effectively captures long-term dependencies and improves computational efficiency.
(3) PSO ensures optimal hyperparameter configurations, reducing the need for manual tuning and improving predictive performance.
(4) The proposed approach is validated using two durability datasets. Results confirm that the proposed approach outperforms baseline TCN, CNN, and RNN models, achieving superior prediction accuracy and robustness.

The remainder of the paper is arranged as follows: Section 2 introduces the principle of the baseline TCN. Section 3 details the proposed RF-TCN prognostic approach. Section 4 presents the validation, comparison, and analysis of the proposed approach. Finally, Section 5 concludes the paper.

2. Temporal convolutional network

The temporal convolutional network (TCN), developed by Bai et al. [46], is designed for time sequence modelling tasks. As presented in Fig. 1, the TCN architecture comprises causal convolution, dilated convolution, and residual blocks.

2.1. Dilated causal convolution

In basic causal convolutional networks, a 1D fully convolutional network with zero padding is employed to ensure that the length of each layer aligns with its preceding layer, as illustrated in Fig. 1(a). Additionally, causal convolutions are utilized to restrict operations at time t to inputs up to and including time t, effectively preventing future information leakage. However, tasks involving long historical data require deeper networks or larger filters, as the ability to retain historical information grows linearly with network depth and filter size. This results in greater model complexity and higher computational costs.

To overcome this challenge, dilated convolution layers are introduced as an enhancement to causal convolution layers. By exponentially expanding the receptive field with network depth, dilated convolutions enable the efficient capture of long-term historical correlations without requiring excessively deep architectures [6]. For a 1D sequence of an

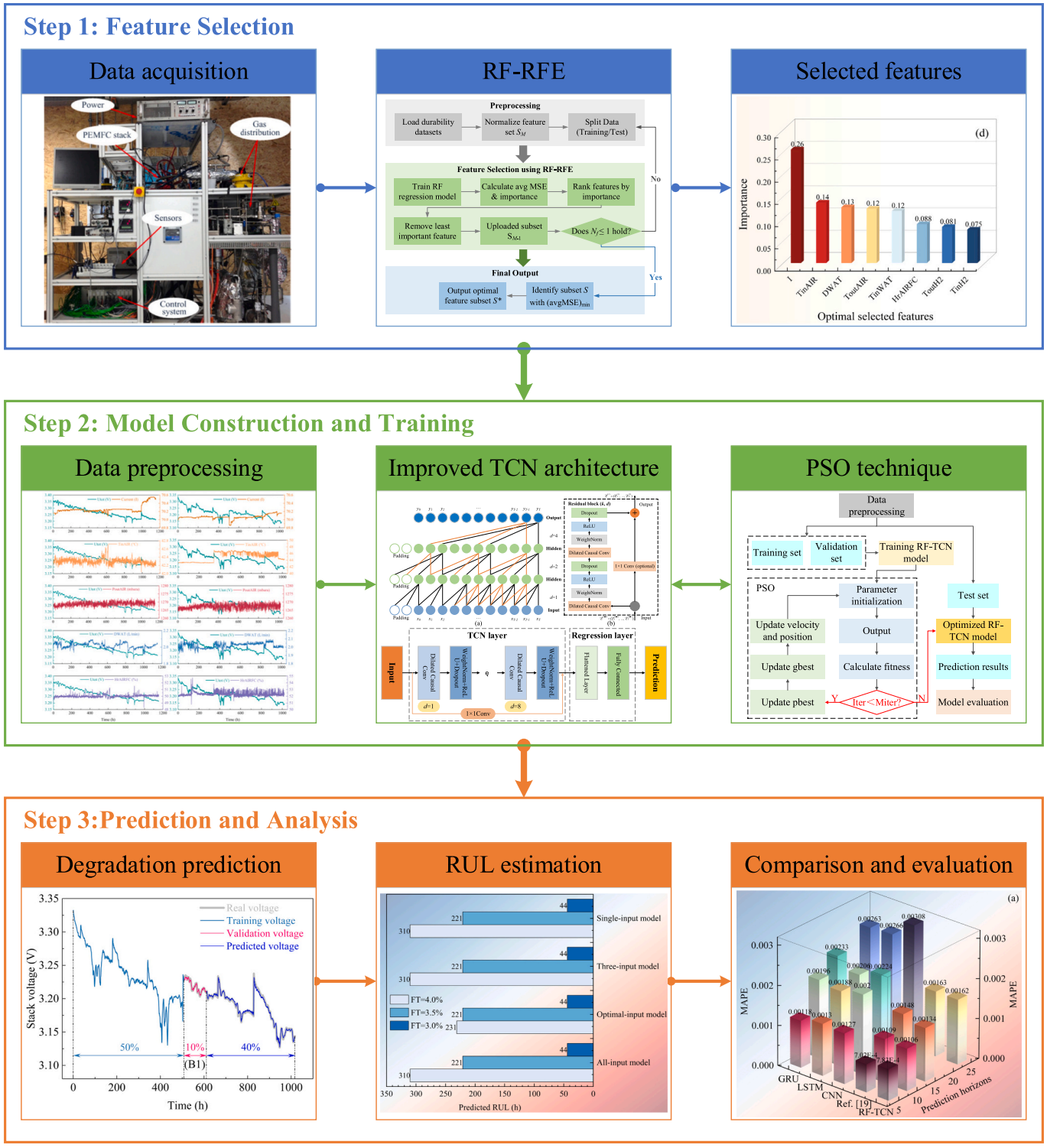


Fig. 2. The framework of the RF-TCN prognostic approach.

input $x \in R^n$ and a filter $f : \{0, \ldots, k-1\} \rightarrow R$, the dilated convolution operation F on the sequence element s is described as [46].

$$F(s) = (x *_d f)(s) = \sum_{i=0}^{k-1} f(i) \cdot x_{s-d \cdot i} \tag{1}$$

where * represents the convolution operator, d denotes the dilation factor, k means the filter size, $x_{s-d \cdot i}$ indicates the $(s - d \cdot i) - th$ element within input sequence, and $s - d \cdot i$ signifies the past direction.

As an example of Fig. 1(a), a three-layer dilated causal convolution with filter size 3 and dilation factors of 1, 2, and 4 demonstrates how these architectures span the entire input sequence. The dilation factor typically follows an exponential progression with base 2, defined as:

$$d = 2^{l-1} \tag{2}$$

where l is the level of the network ($l = 1, 2, \ldots, l$).

When the dilation factor (d) is 1, the dilated causal convolution

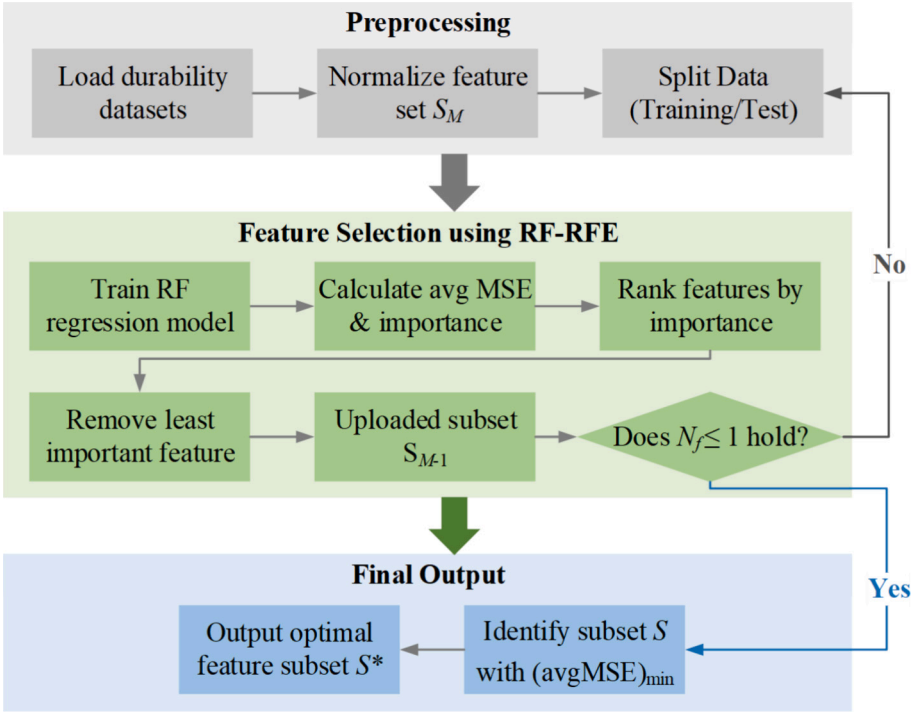


Fig. 3. The implementation process of RF-RFE for selecting the optimal feature subsets.

reduces to an ordinary one. Utilizing larger dilation factors enables networks to cover longer historical inputs, thereby effectively increasing the receptive field of convolutional networks while significantly reducing network depth.

2.2. Residual connection

The residual network effectively addresses the gradient explosion and vanishing in deep architectures [48]. A residual block introduces a sequence of transformations $F(x)$ added to the block's input x, allowing layers to adjust the identity mapping instead of performing full transformations [46]. The formulation is given by:

$$y = Activation(x + F(x)) \tag{3}$$

where y is the output, $Activation(\cdot)$ is the activation function, x is the input, and $F(x)$ is the output of the convolution layer.

As depicted in Fig. 1 (b), a residual block includes two layers of dilated causal convolutions, weight normalization, Rectified Linear Units (ReLU), and dropout. The two middle components help accelerate training and enhance generalization [6], while the last component serves to regularize the model [53]. Additionally, a 1×1 convolution ensures that the identity mapping channel has the same width as the output [46], as illustrated in Fig. 1(c).

3. Proposed RF-TCN prognostic approach

As illustrated in Fig. 2, the RF-TCN prognostic approach comprises three sequential phases: feature selection, model construction and training, and model prediction and analysis.

In the first phase, the RF-RFE method is applied to select the features related to degradation from experimental monitoring variables. This iterative approach eliminates the least significant features in each step, ultimately producing an optimal feature subset. The second phase focuses on constructing and training a predictive model using an improved TCN architecture. Two improvement strategies are integrated to refine the baseline TCN, and the model is trained with the feature subset obtained in the first phase. Additionally, the PSO algorithm is utilized to configure the model's hyperparameters, ensuring optimal performance. In the final phase, the trained predictive model is employed to forecast PEMFC degradation and estimate RUL. The model's predictive accuracy is validated against real-world data, and its performance is compared with traditional deep learning models to demonstrate the advantages of the proposed approach.

3.1. Feature selection

3.1.1. Data acquisition

Two durability datasets from the IEEE PHM 2014 data challenge, as detailed in [54,55], are utilized to verify the proposed approach. Detailed descriptions of these datasets are provided in subsection 4.1.

3.1.2. Feature importance assessment based on RF

The RF algorithm is an ensemble technique that integrates multiple decision trees to enhance the predictive precision and control model overfitting [42,56]. Known for its stability and ability to visualize variable importance, RF is widely employed for selecting features in data-driven models [57]. Each tree is trained on a randomly sampled sub-dataset, and the final prediction is obtained by averaging individual tree output [58].

Feature importance is assessed through the Permutation Importance score, calculated by comparing the prediction accuracy before and after introducing noise to out-of-bag (OOB) data [58,59], as follows.

$$\overline{I}_i = \frac{1}{B}\sum_{i=1}^{B}\left(E_{bi}^{OOB} - E_b^{OOB}\right) \tag{4}$$

where $\overline{I}_i$ represnts the importance of each feature x_i, B means the number of decision trees, E_b^{OOB} denotes the number of classification errors of OOB data in individual sub-decision trees, and E_{bi}^{OOB} signifies the number of classification errors when randomly permuting the value of OOB data x_i of individual decision trees. It is worth noting that a substantial increase in OOB error following noise introduction indicates that the corresponding feature has a strong influence on the model's predictive performance [58].

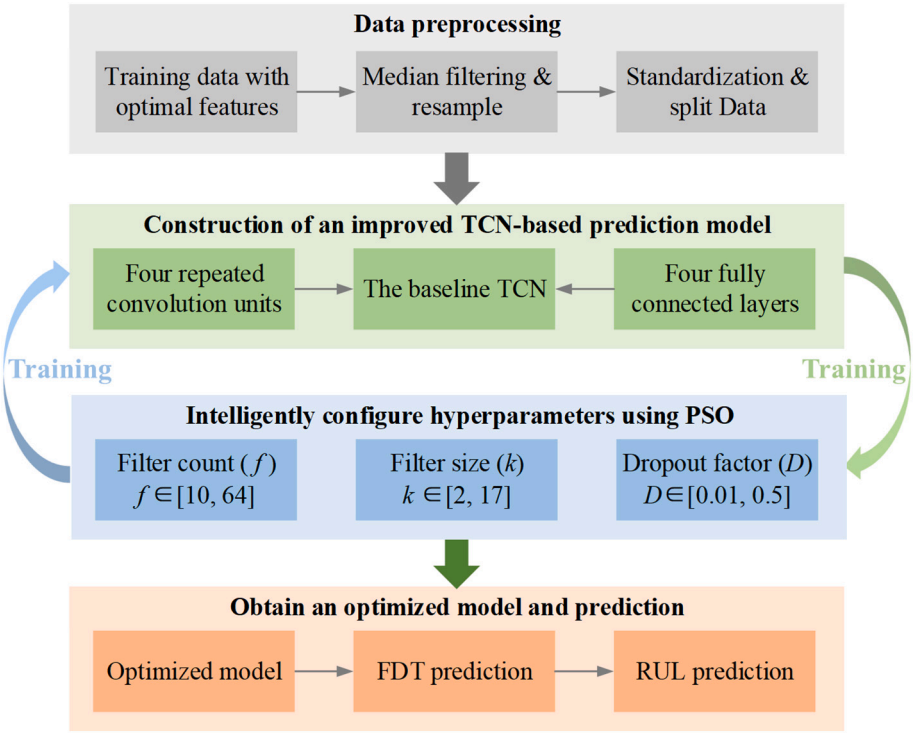


Fig. 4. The flowchart of model construction and training process.

3.1.3. Feature selection using RF-RFE

Feature selection aims to identify the most informative features from the original feature space, improving the effectiveness and robustness of the predictors [36]. The RF-RFE method leverages RF's ability to handle complex nonlinear relationships and RFE's iterative elimination of unimportant features.

It is expected to reduce data redundancy and produce a more compact feature set, thereby enhancing the reliability and accuracy of feature selection. The RF-RFE feature selection process is illustrated in Fig. 3.

The implementation process is detailed below:

Step 1: Load and normalize the original feature set S_0 with 18 variables, as described in subsection 4.2, to form the current feature set $S_M = \{s_1, s_2, \ldots, s_M\}$. Define N_f, the number of features in the updated subset $S_{M\text{-}1}$, as the iteration criterion ($N_f \geq 1$). Split the data into 70 % for training and 30 % for testing.

Step 2: Construct the RF regression model using S_M. The number of decision trees is set to 100 [33], and the maximum number of splits uses the default automatic selection option in MATLAB. To improve model performance, apply ten-fold cross-validation during training.

Step 3: Compute the average mean square error (avgMSE) across ten iterations to evaluate model performance [38]. Simultaneously, calculate the average importance score for each feature in S_M over ten iterations and rank the features in descending order of importance.

Step 4: Using the RFE algorithm, remove the least important feature with the lowest average importance score. Update the remaining feature set to $S_{M\text{-}1}$ and construct a new regression model. Repeat Steps 2–4 until $N_f = 1$.

Step 5: Compare the avgMSE values recorded across all feature subsets $S = \{S_M, S_{M\text{-}1}, \ldots, S_1\}$. Identify the subset with the minimum avgMSE as the optimal feature subset S^*, and output S^*.

3.2. Model construction and training

This section describes the development and training process of the prediction model, as presented in Fig. 4. In the data preprocessing stage, median filtering is applied to reduce noise, followed by resampling to ensure adequate data representation. The processed data is standardized and split into training, validation, and testing sets. In the model construction and training stage, an improved prediction model is developed by integrating two enhancement strategies into the baseline TCN architecture, aiming to effectively capture temporal dependencies and represent features. The model is trained using the training set, while PSO is used to intelligently optimize hyperparameters, ensuring optimal performance. The prediction stage utilizes the optimized model to predict the FDT and RUL.

3.2.1. Data preprocessing

Data preprocessing is critical for performing degradation prognostics of PEMFCs as the monitoring dataset usually contains numerous raw data points and significant noise. To reduce computational costs and improve training efficiency, median filtering with a window size of 5 [25] is applied first to eliminate noise and restore the original data pattern. The filtered data are then resampled at half-hour intervals [17,30] to create new degradation datasets [15]. Before training the model, the resampled datasets are standardized using Eq. (5a) [23] and segregated into training, validation, and testing sets in specific proportions. Notably, the validation set helps to assess the model's performance independently of the training and testing data [22].

$$x_i = \frac{X_i - X_{mean}}{X_{std}} \tag{5a}$$

$$X_{mean} = \frac{1}{N}\sum_{i=1}^{N} X_i \tag{5b}$$

$$X_{std} = \sqrt{\frac{\sum_{i=1}^{N}(X_i - X_{mean})^2}{N}} \tag{5c}$$

where x_i means the standardized data, X_i indicates the input data, X_{mean} represents the mean of the dataset, and X_{std} denotes the standard devi-

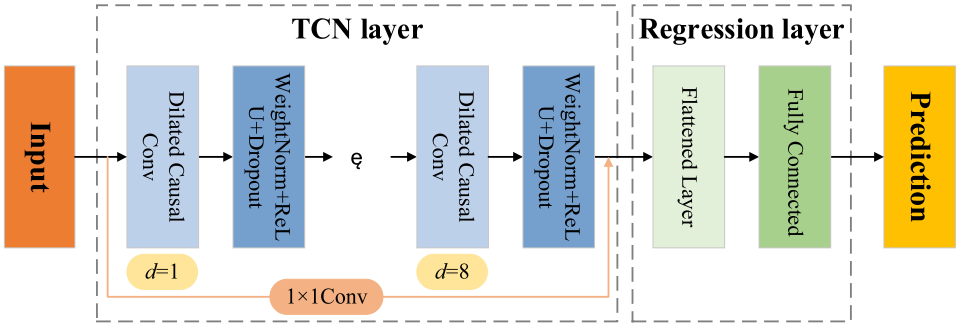


Fig. 5. Improved TCN-based predictive model for PEMFC.

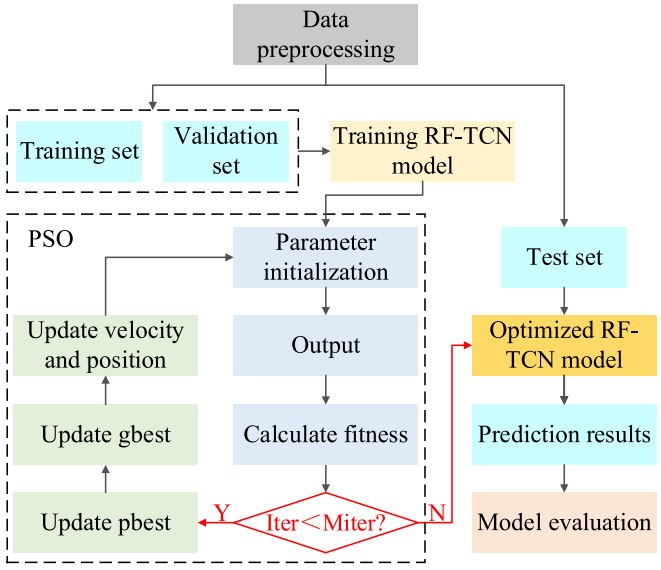


Fig. 6. Optimization process using the PSO algorithm for the prediction model.

ation of the dataset.

3.2.2. New improved TCN-based prediction model

As described in subsection 2.1, the TCN architecture excels in handling time series prediction tasks. However, multivariate predictive models with higher dimensions require a wider receptive field to effectively capture long-term dependencies. This optimization involves adjusting the convolution filter size (k), dilation factor (d), and network depth (n), while considering computational costs [48]. To enhance the prediction performance of the baseline TCN architecture, two modification strategies are implemented:

(1) Four repeated convolution units are integrated into the original residual block structures to enable multi-layer feature extraction and accurate representation of local degradation properties. This approach improves feature extraction capabilities while avoiding the higher computational costs associated with stacking multiple residual blocks [48,49].
(2) A flattened layer and four fully connected layers are introduced in the regression stage to fuse extracted local features, providing a more comprehensive representation of PEMFC degradation.

The resulting improved TCN-based predictive model is presented in Fig. 5. The model comprises an input layer, a TCN layer, and a regression layer. Each repeated unit in the TCN layer is structured as *Dilated Causal Conv* + *WeightNorm* + *ReLU* + *Dropout*, with all convolution layers sharing the same filter size and a stride of one [6]. The dilation factors (d) for successive layers are set to 1, 2, 4, and 8, while dropout is applied only in the TCN layer. In the regression layer, the neuron counts for four fully connected layers are configured to 48, 32, 16, and 1, respectively [6].

3.2.3. Optimization technique

The PSO optimization process for the prediction model is illustrated in Fig. 6. First, a set of initial parameters for the prediction model is provided. Second, the PSO algorithm iteratively optimizes the hyperparameters to minimize the loss function value (fitness), which quantifies the discrepancy between the target output and the predicted output on the validation sets [57]. This training process is repeated until the maximum number of iterations is reached. The trained model is applied to PEMFC prognostics, and its predictive performance is assessed using test sets.

The hyperparameters of the developed prediction model include the filter count (f), filter size (k), the number of residual blocks (n), and the dropout factor (D), where their search ranges are determined according to the results of existing literature as follows: $f \in [10, 64]$ [16,19], $k \in [2, 17]$ [6,16,19], $n \in [1, 1]$, and $D \in [0.01, 0.5]$ [22,46]. Notably, the performance enhancement of the TCN is achieved by stacking multilayer convolution units in the original residual block structure; thus, the number of residual blocks n is kept at 1. The hyperparameter optimization process is implemented using MATLAB software, with the initial parameters specified as follows [25,51,52]: a population size of 20, a maximum number of iterations of 50, a minimum inertia weight of 0.4, a maximum inertia weight of 0.9, personal and global learning coefficients both set to 2 and a maximum particle velocity of 6.

3.2.4. Model evaluation metrics

Commonly used model evaluation metrics, such as MAE, mean absolute percentage error (MAPE), RMSE, and the coefficient of determination (R^2), are employed to assess the predictive model's performance [19,57]. The expressions are given by:

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_t - p_t| \tag{6}$$

$$\mathrm{MAPE} = \frac{1}{N}\sum_{i=1}^{N}\frac{|y_t - p_t|}{|y_t|} \times 100\% \tag{7}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_t - p_t)^2} \tag{8}$$

$$R^2 = 1 - \sum_{i=1}^{N}(y_t - p_t)^2 \Big/ \sum_{i=1}^{N}(y_t - \overline{y_t})^2 \tag{9}$$

where y_t and p_t denote the real and predicted voltages of the fuel cell stack, respectively. N indicates the total sample size. $\overline{y_t}$ is the average value of the real voltages.

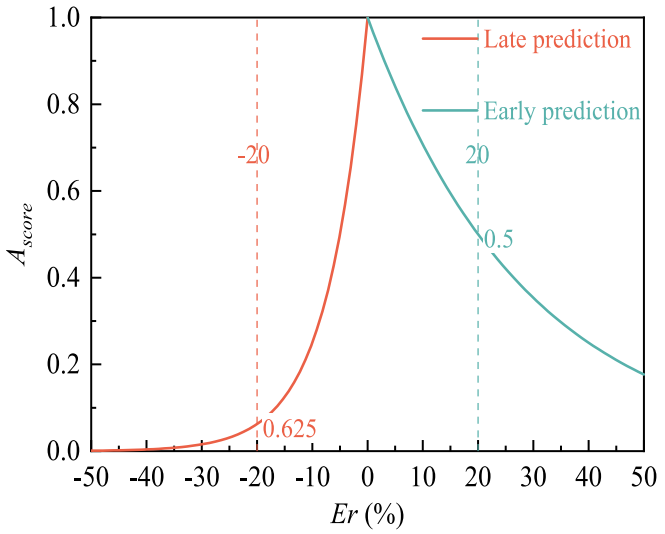


Fig. 7. Scoring function of a RUL estimate based on per cent error.

Smaller values of MAE, MAPE, and RMSE signify higher prediction accuracy, while R^2 reflects the degree of correlation between the model outputs and targets. An R^2 value close to 1 represents a perfect fit of the model's output relative to input data [60].

3.3. Prediction and analysis

3.3.1. Degradation trend prediction method

PEMFC degradation trend prediction is a nonlinear time series task that requires constructing input-output patterns in a time series dataset [22]. This study employs the multi-step-ahead prediction method described in [19,23] to produce input-output data pairs for training, validation, and testing sets [25]. The method predicts several steps ahead by learning from earlier values, and after each prediction, the actual measurements are fed into the model [19]. This approach offers at least two advantages: maintaining a close input-output relationship and avoiding cumulative prediction errors [50].

3.3.2. RUL calculation and evaluation indicator

The RUL is defined as the duration until a specified voltage loss occurs [9]. In this study, the failure threshold (FT) for FC1 is set at 3.0 %, 3.5 %, and 4.0 % of the initial stack voltage ($V_{ini} = 3.3482\ V$) [17,30]. Similarly, the FT for FC2 is set at 4.0 %, 4.5 %, and 5.0 % of the initial stack voltage ($V_{ini} = 3.317\ V$) [17,30].

To assess the model's performance in predicting RUL, the final score of the RUL prediction accuracy (FA_{score}) is calculated as follows:

(1) The real RUL (O_{RUL}) and predicted RUL (P_{RUL}) are expressed as:

$$O_{RUL} = t_{rFT} - t_{pre} \tag{10}$$

$$P_{RUL} = t_{pFT} - t_{pre} \tag{11}$$

where t_{pre} represents the initial prediction time, t_{rFT} denotes the first time when the real voltage reaches FT, and t_{pFT} means the first time when the predicted voltage reaches FT.

(2) The per cent error (E_r) is defined as:

$$E_r = \frac{O_{RUL} - P_{RUL}}{O_{RUL}} \times 100\% \tag{12}$$

(3) According to [54], the score of the RUL prediction accuracy (A_{score}) is given by:

$$A_{score} = \begin{cases} exp(-ln(0.5) \times 0.2E_r); & E_r \le 0 \\ exp(+ln(0.5) \times 0.05E_r); & E_r > 0 \end{cases} \tag{13}$$

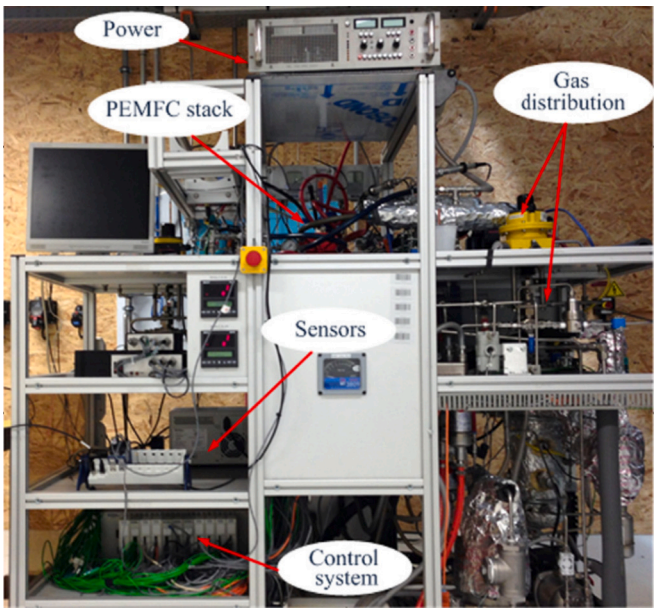


Fig. 8. PEMFC system experiment platform [55].

Table 2
Range of controlled physical parameters [55].

Parameters	Control range	Parameters	Control range
Cooling temperature	20–80 [°C]	Air flow	0–100 [1/min]
Cooling flow	0–10 [l/min]	H_2 flow	0–30 [1/min]
Gas temperature	20–80 [°C]	Gas pressure	0–2 [bara]
Gas humidification	0–100 [%]	Load current	0–300 [A]

Table 3
Monitoring operational data in the test [55].

Index	Physical meaning (unit)
U1 to U5, Utot	Single-cell and stack voltage (V)
I, J	Current and (A) and current density (A/cm^2)
TinH2, ToutH2	Temperature of H_2 at the inlet and outlet (°C)
TinAIR, ToutAIR	Temperature of Air at the inlet and outlet (°C)
TinWAT, ToutWAT	Temperature of cooling water at the inlet and outlet (°C)
PinH2, PoutH2	Pressure of H_2 at the inlet and outlet (mbara)
PinAIR, PoutAIR	Pressure of Air at the inlet and outlet (mbara)
DinH2, DoutH2	Flow rate of H_2 at the inlet and outlet (L/min)
DinAIR, DoutAIR	Flow rate of Air at the inlet and outlet (L/min)
DWAT	Flow rate of cooling water (L/min)
HrAIRFC	Hygrometry of air at the inlet (%)

In Eq. (13), $E_r \le 0$ represents overestimation predictions with a severe deduction, while the $E_r > 0$ represents underestimation predictions with a slight deduction, as shown in Fig. 7.

(4) The average score (FA_{score}) for different FTs is computed as:

$$FA_{score} = \frac{1}{Z}\sum_{i=1}^{Z}(A_{score}) \tag{14}$$

where Z denotes the number of selected FTs, which is 3 both for FC1 and FC2.

4. Experimental verification

4.1. Durability dataset description

The PEMFC ageing durability dataset provided by the FCLAB Research Federation [54] is utilized to validate the efficacy and

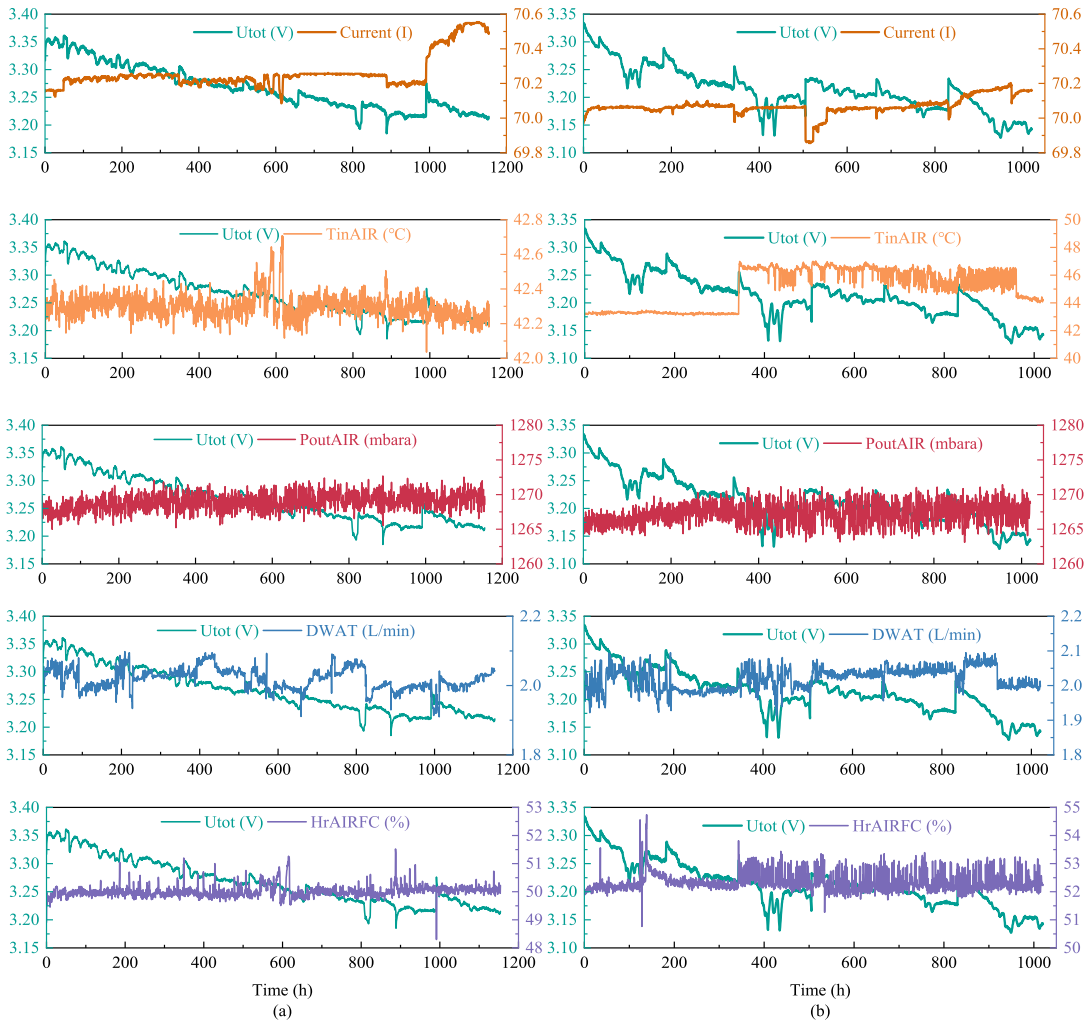


Fig. 9. A portion of reconstructed degradation data of (a) FC1 and (b) FC2.

advantages of the proposed approach. The test platform, illustrated in Fig. 8, operates under controlled physical parameters presented in Table 2. Ageing tests are conducted under two conditions: static loading (FC1) and quasi-dynamic loading (FC2), with durations of 1154 h and 1020 h, respectively.

The FC1 test is performed with a constant current of 70 A, while the FC2 test involves a dynamic current of 70 A with 7 A ripples at a frequency of 5 kHz. The platform records 24 operational variables, as described in Table 3, where U1 to U5 and Utot are the performance output variables, while the rest are operational parameters. Of these, the stack voltage, which gradually decreases over time, reflects the degradation behaviour of the PEMFC stack to some extent [30,61] and is

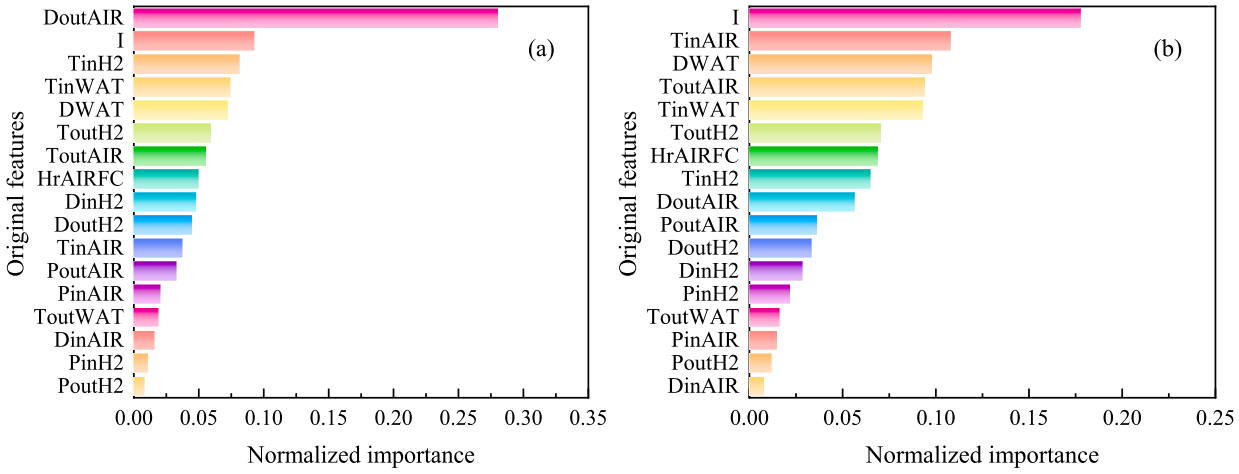


Fig. 10. Normalized importance of all features: (a) FC1; (b) FC2.

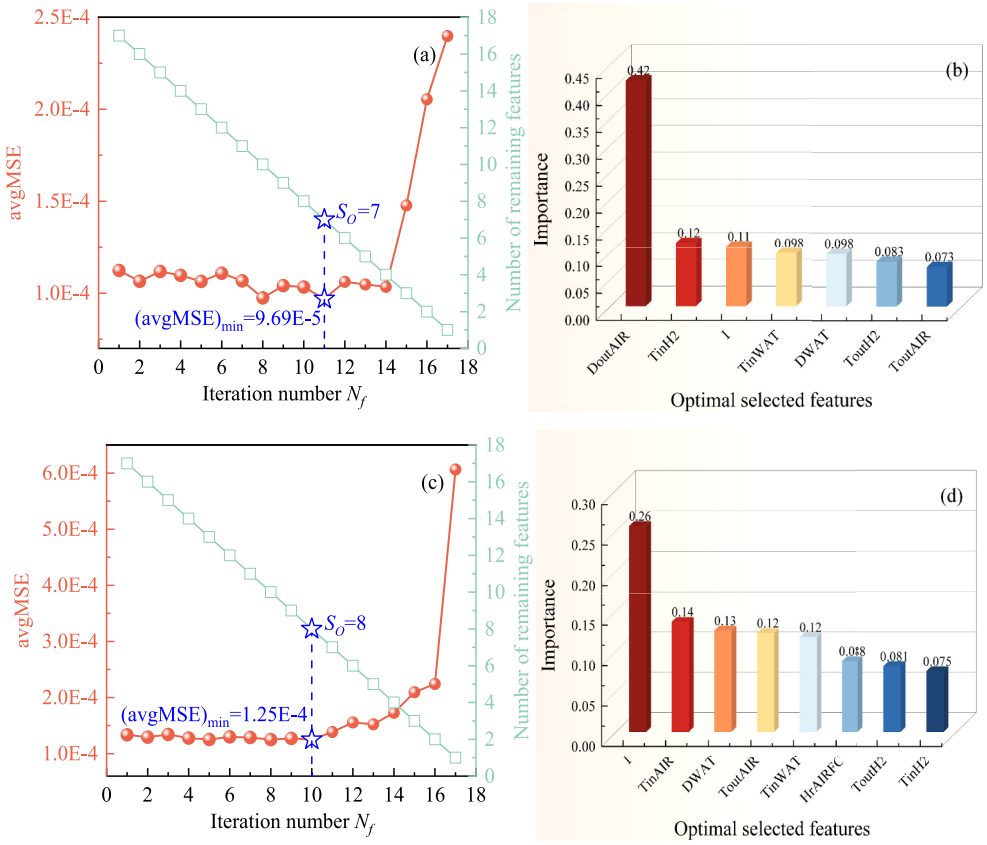


Fig. 11. Feature selection results: (a) and (c) show the relationship between the iteration number, the number of remaining features, and the corresponding avgMSE for FC1 and FC2, respectively. (b) and (d) illustrate the importance of optimal features for FC1 and FC2, respectively.

widely accepted as a health performance indicator [17]. Therefore, the stack voltage is selected as the output variable for regression analysis. A portion of the reconstructed degradation data for FC1 and FC2, processed as described in subsection 3.2.1, is presented in Fig. 9. Each result includes the stack voltage and a corresponding operational variable from the test.

The proposed approach is developed in MATLAB 2023b. The predictive model is trained using the Adam optimizer with an initial learning rate of 0.001, a training epoch of 120, and a mini-batch size of 120. Hyperparameter configuration is achieved using PSO algorithms, as detailed in subsection 3.2.3. All experiments are conducted on a system equipped with an Intel i9-12900H CPU, 16GB RAM, and a GeForce RTX 3060 GPU with 6GB of memory.

4.2. Feature selection result and analysis

In this section, the RF-RFE method described in subsection 3.1.3 is utilized to identify optimal features from the monitoring operational variables, which serve as inputs for the deep learning model. Notably, individual cell voltage data (U1 to U5) are excluded from the original feature set (S_0) because they are difficult to obtain in practical applications. Similarly, the variable (J) is excluded, as the current variable (I) not only reflects actual load fluctuations but also indicates changes in the cell's active area. Ultimately, the remaining 18 monitoring operational variables form the original feature set S_0, with the stack voltage (Utot) used as the output variable.

Fig. 10 shows the normalized importance of all features during the first iteration. For the same variable, the importance score ranking differs under different current loading conditions, aligning with findings in [17,27]. This discrepancy can be attributed to the statistical

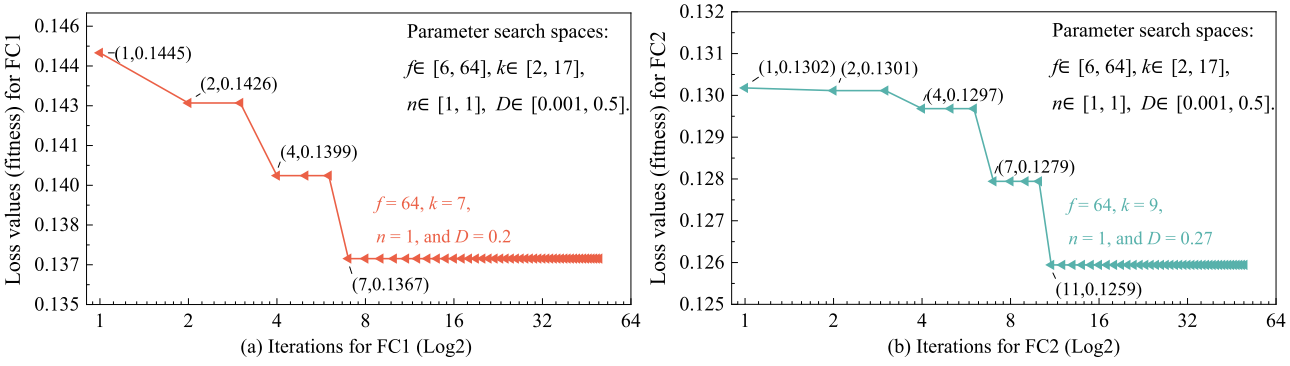


Fig. 12. The convergence plots of the hyperparameter optimization process for (a) FC1 and (b) FC2.

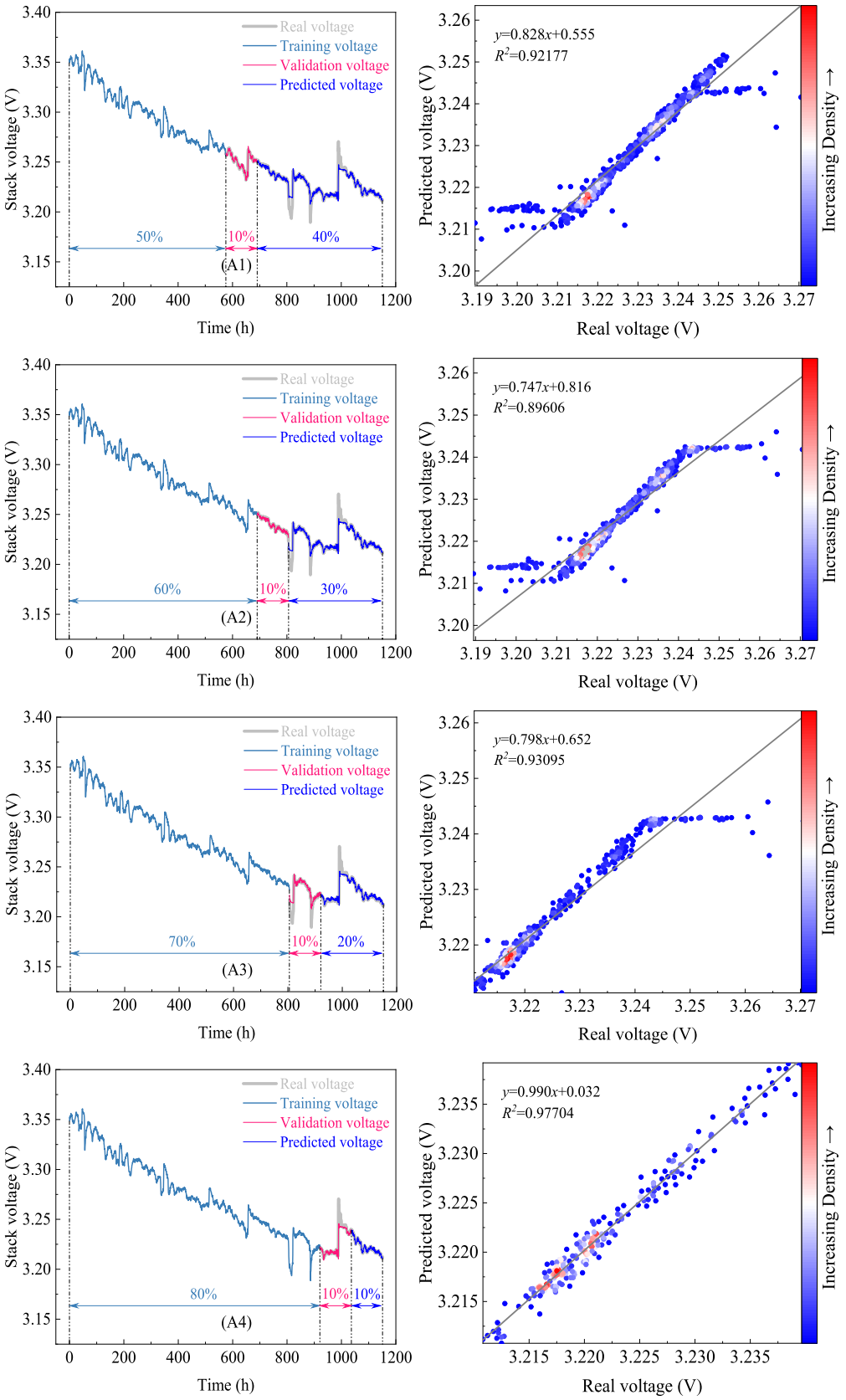


Fig. 13. Predicted stack voltage curves (left) and scatter density plots (right) for FC1 under different training lengths.

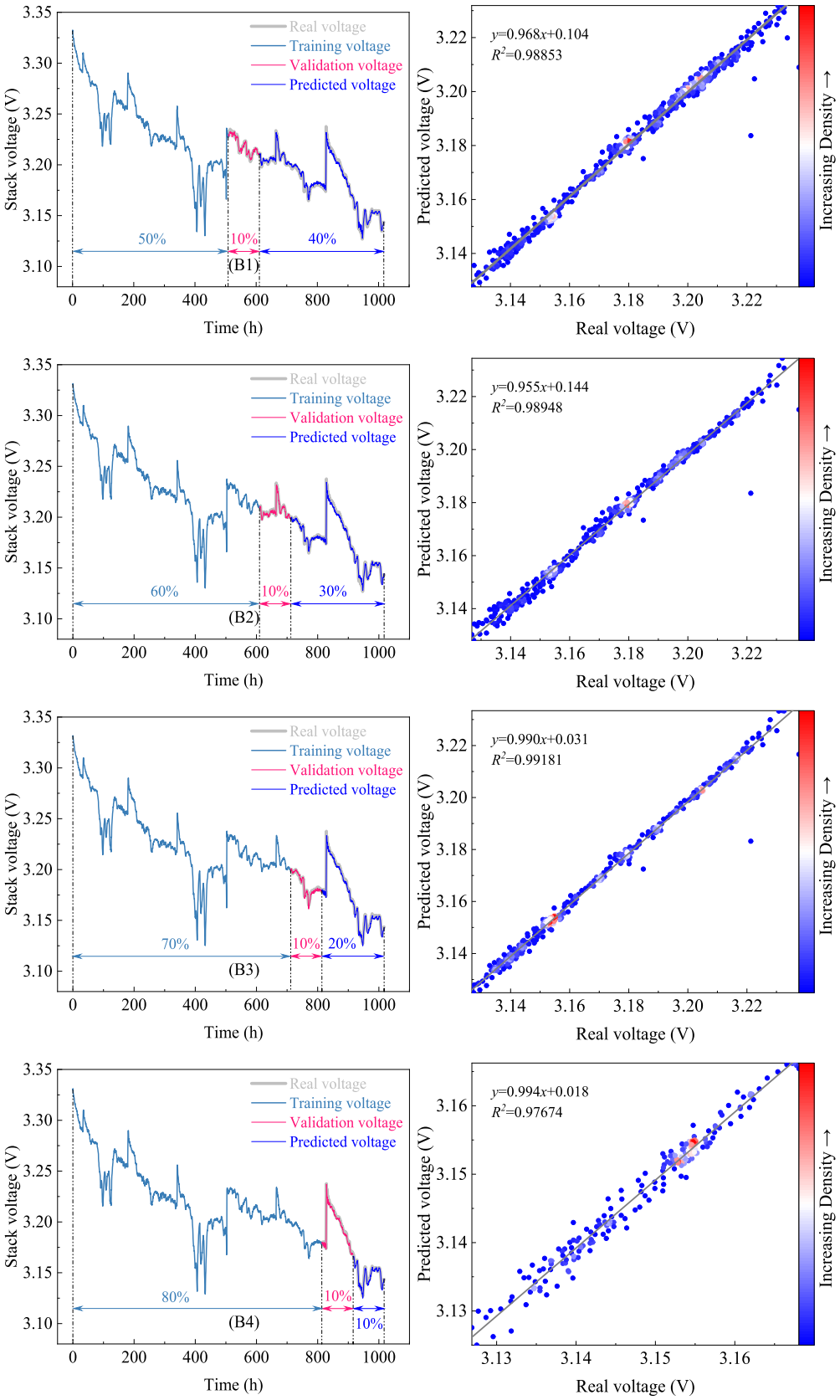


Fig. 14. Predicted stack voltage curves (left) and scatter density plots (right) for FC2 under different training lengths.

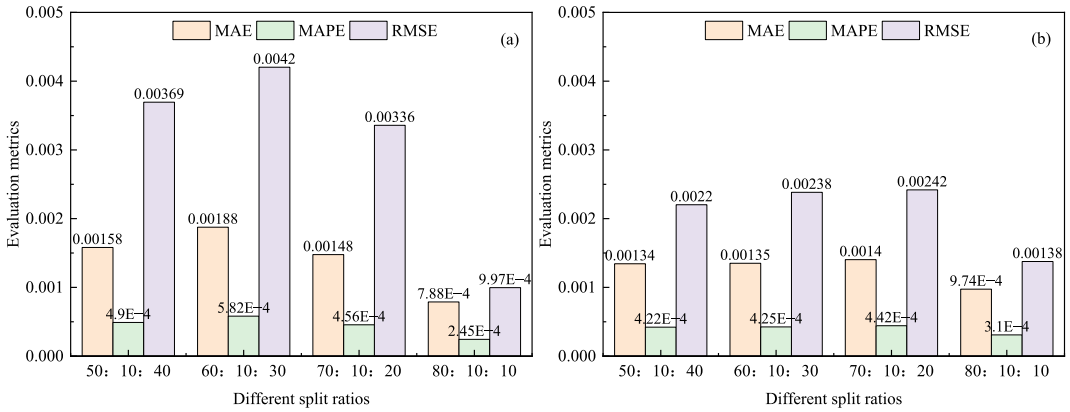


Fig. 15. Performance metrics of the proposed model using different split ratios for (a) FC1 and (b) FC2.

characteristics of the datasets themselves [62].

The feature selection results for FC1 and FC2 are presented in Fig. 11. From Fig. 11(a) and (c), the iteration process is negatively correlated with the number of remaining features. As iterations progress, the avgMSE initially stabilizes at a lower value, reaches a critical turning point (N_f), and subsequently increases. In comparison to the results in [38] (Fig. 6-4a and 6-5a in [38]), the RF-RFE generates a smoother curve, indicating enhanced stability. Based on the principle of minimal avgMSE, the number of optimal features in the subset S_O is determined to be 7 for FC1 and 8 for FC2. The selected features are presented in Fig. 11(b) and 11(d), and the results are generally consistent with the mechanistic analyses in [4,63].

The feature selection approach of this study offers two significant advantages over previous methods. First, the feature selection process is automated through an iterative RF-RFE algorithm, eliminating the need for manual, assumption-based selection [28,30]. Second, by excluding performance output variables such as U1 to U5, the method maximizes the use of operational variables, creating a more compact feature subset than those in [17]. Based on these benefits, the selected features are expected to enhance model performance, particularly in terms of reducing feature space dimensionality and computational cost. Specifically, the optimal features selected via RF-RFE are anticipated to contribute to a more accurate and comprehensive description of PEMFC performance degradation.

4.3. Hyperparameter optimization results and analysis

The hyperparameter optimization process is illustrated in Fig. 12, where lower fitness values indicate improved model performance. The convergence plots for both the FC1 and FC2 datasets demonstrate steady and smooth loss curves, with loss values gradually decreasing and converging within 11 iterations, well below the preset maximum of 50. This rapid convergence highlights the efficiency of the PSO algorithm in minimizing model error while optimizing hyperparameters. The

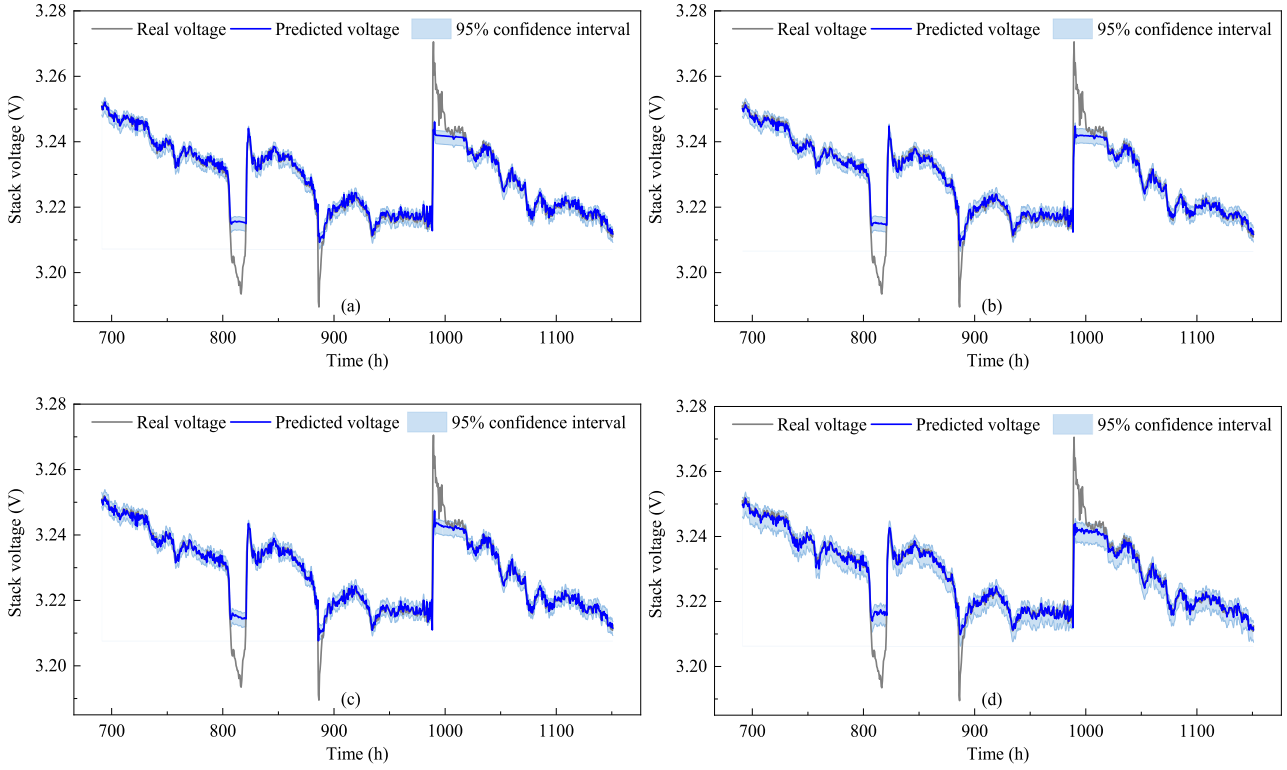


Fig. 16. Predicted results for FC1 using four different input feature sets: (a) single-input model, (b) three-input model, (c) optimal-input model, and (d) all-input model.

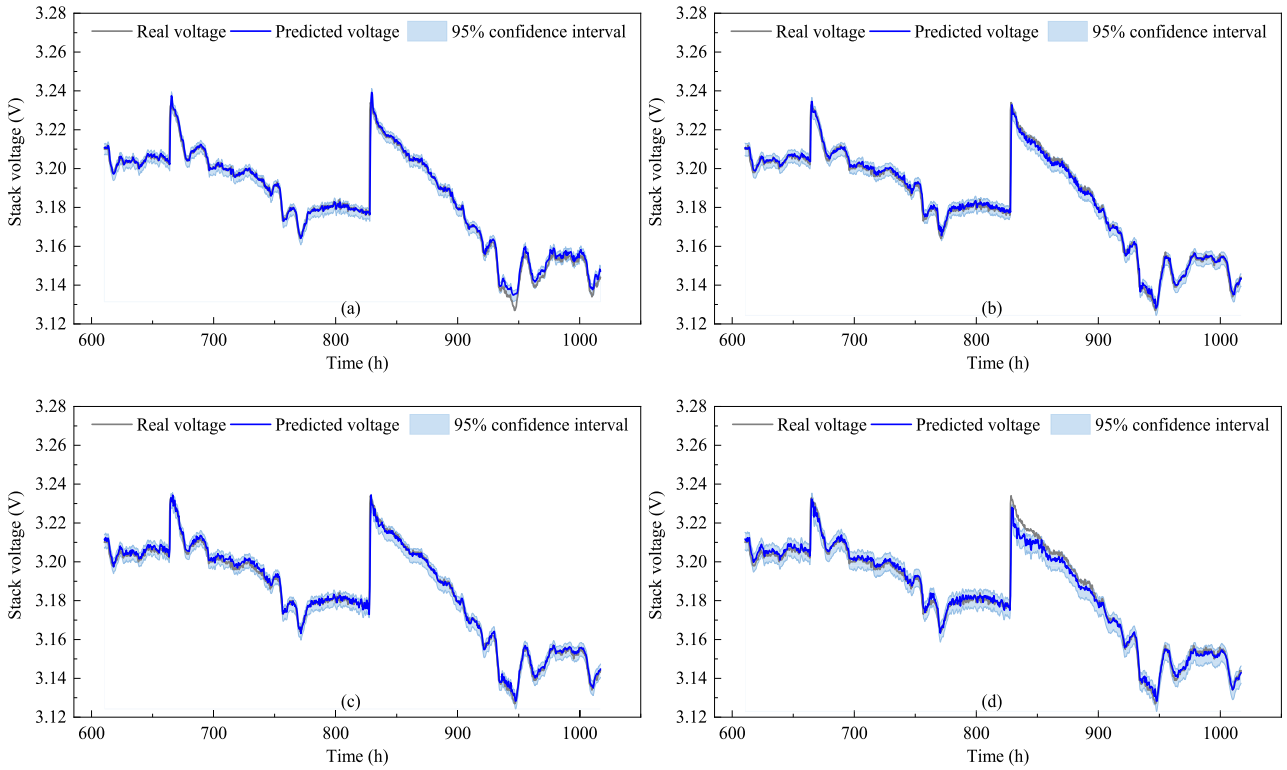


Fig. 17. Predicted results for FC2 using four different input feature sets: (a) single-input model, (b) three-input model, (c) optimal-input model, and (d) all-input model.

optimized hyperparameters are as follows: $f = 64$, $k = 7$, $n = 1$, and $D = 0.2$ for FC1, and $f = 64$, $k = 9$, $n = 1$, and $D = 0.27$ for FC2. These parameters will be used in subsequent analyses to ensure robust performance.

4.4. Results and discussion

This section presents the prediction results of the proposed model under different training lengths and input feature sets. Additionally, three commonly used models, including CNN, LSTM and GRU models, along with the baseline TCN model [46] and results from the literature [19], are selected for comparison. All models are evaluated using the same dataset and prognostic method described in subsection 3.3.1.

4.4.1. FDT prediction results under different training lengths

To evaluate the model's generalization ability, the prediction performance of the proposed model is tested using training dataset lengths of 50 %, 60 %, 70 %, and 80 % of the total samples. The validation set lengths are kept at 10 %, with the remaining data as test sets.

Fig. 13 and Fig. 14 present the predicted stack voltage curves and corresponding scatter density plots for FC1 and FC2 under different training lengths. The results show that with varying training lengths, the predicted voltage curves effectively follow the real voltage reduction and recovery trends. In the scatter density plots, the X-axis denotes the real voltage, while the Y-axis signifies the predicted voltage through the proposed model. The grey straight line represents the points where predicted voltages coincide with real voltages. The nearer these scatter points are to the grey line, the more accurate the prediction. It can be observed that across all training lengths, the scatter density plots exhibit a high consistency agreement between predicted and real voltages.

Fig. 15 depicts the performance evaluation metrics (MAE, MAPE, and RMSE) for the proposed model across different training lengths. It is clear that with varying training lengths, the maximum values of these metrics for both FC1 and FC2 remain low and are less than the best prediction results in [38] (Fig. 3–9 and Fig. 3–10 in [38]). These results demonstrate that the proposed model achieves high prediction accuracy and exhibits robust generalization, with its performance minimally impacted by variations in training length.

4.4.2. FDT prediction results under different input feature sets

To assess the efficacy of the optimal features selected via the RF-RFE method in predicting the FDT of PEMFC, 50 % of the sample dataset is utilized for training, and the impact of input features on prediction performance is examined for FC1 and FC2. Four models with different

Table 4
Comparison of prediction performance for FC1 and FC2 using four models with different input feature sets. The best results are marked in **bold**, and the second-best results are in blue.

Datasets	Models	MAE	MAPE	RMSE	PICP	PINAW	AIS
FC1	Single-input model	0.001705	0.0005290	0.003851	0.8643	**0.04978**	0.003539
	Three-input model	0.001663	0.0005160	0.003816	0.8903	0.05397	0.003431
	Optimal-input model	**0.001581**	**0.0004900**	**0.003694**	**0.9001**	0.05338	**0.003294**
	All-input model	0.001816	0.0005630	0.004050	0.8817	0.07427	0.003519
FC2	Single-input model	0.001916	0.0006036	0.003190	0.8690	0.06220	0.001645
	Three-input model	0.001557	0.0004888	0.002816	0.9103	**0.05964**	0.001410
	Optimal-input model	**0.00134**	**0.0004218**	**0.002202**	**0.9730**	0.06249	**0.001094**
	All-input model	0.002060	0.0006450	0.003180	0.8894	0.08092	0.002177

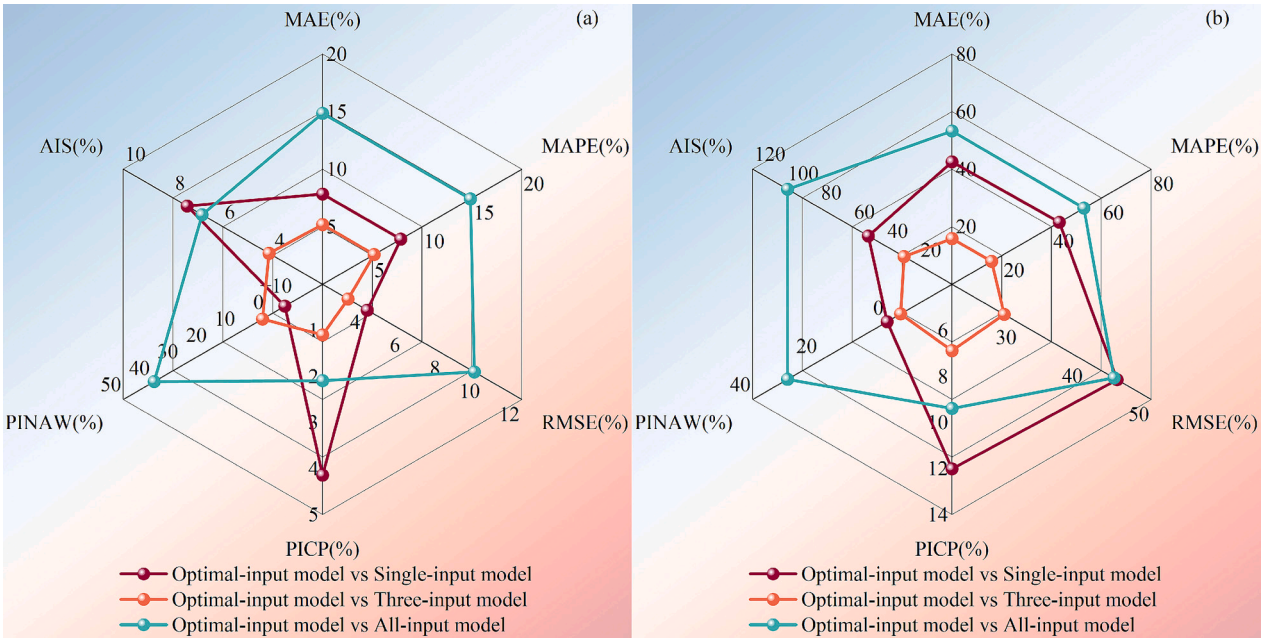


Fig. 18. Comparative improvement analysis in evaluation metrics for the optimal-input model relative to other input models: (a) FC1, and (b) FC2.

input feature sets are designed for comparison:

(a) Single-input model: Uses the stack voltage (Utot) as the sole input, serving as a reference.
(b) Three-input model: Incorporates the two most important variables (Fig. 11) in addition to Utot. TinH2, DoutAIR, Utot for FC1, and I, TinAIR, Utot for FC2.
(c) Optimal-input model: Utilizes the optimal feature sets described in Fig. 11.
(d) All-input model: Involves all 18 initial input features.

The prediction results for FC1 and FC2 are displayed in Fig. 16 and Fig. 17, respectively, with a detailed comparison of performance metrics given in Table 4. To assess the uncertainty of prediction results, kernel density estimation [64] is employed. Metrics related to prediction interval coverage probability (PICP), prediction interval normalized average width (PINAW), and average interval score (AIS) are included, with details available in Ref. [38].

From Fig. 16 and Table 4, the following observations can be made: (i) The optimal-input model yields the best prediction accuracy, with the smallest MAE, MAPE, and RMSE of 0.001581, 0.0004900, and 0.003694, respectively. (ii) The optimal-input model provides the highest prediction interval quality, with the largest PICP of 0.9001, the second smallest PINAW of 0.05338, and the smallest AIS of 0.003290. (iii) The three-input model ranks second in performance. (iv) The single-input model performs poorly, while the all-input model shows the worst results due to feature redundancy affecting prediction accuracy.

Similar conclusions can be drawn from Fig. 17 and Table 4: (i) The optimal-input model achieves the best prediction accuracy, with the smallest MAE, MAPE, and RMSE of 0.001344, 0.0004218, and 0.002202, respectively. (ii) The optimal-input model shows the highest prediction interval quality, with a PICP of 0.9730, the relatively smallest PINAW of 0.06249, and the smallest AIS of 0.001094. (iii) The three-input model ranks second, while the single-input model performs poorly, and the all-input model shows the weakest performance across most metrics. These results demonstrate that the optimal features selected by the RF-RFE method are effective.

Fig. 18 shows the improvements in the evaluation metrics of the model after using the optimal input feature set for FC1 and FC2. The prediction performance shows varying degrees of enhancement when employing optimal input feature sets for training, with significant improvements in MAE, MAPE, RMSE, PICP, and AIS. For instance, in FC2, compared to the single-input model, the optimal-input model reduces MAE by 42.53 %, MAPE by 43.10 %, RMSE by 44.86 %, improves PICP by 12.02 % and reduces AIS by 50.35 %. Relative to the all-input model, the optimal-input model reduces MAE by 53.24 %, MAPE by 52.93 %, RMSE by 44.39 %, improves PICP by 9.40 %, and reduces AIS by 99.02 %. These results prove that the RF-RFE method effectively selects optimal input features that accurately characterize fuel cell degradation, thereby improving prediction accuracy and reducing uncertainty.

Table 5
Comparison of RUL between real observations and prediction results obtained from models with different input feature sets, along with the corresponding final scores for FC1.

Models	RUL results (h)			Final score (*FAscore*)
	FT = 3.0 %	FT = 3.5 %	FT = 4.0 %	
Real observations	43.5	220.5	230.5	–
Single-input model	44	221	310	0.61
Three-input model	44	221	310	0.61
Optimal-input model	**44**	**220.5**	**231**	**0.94**
All-input model	44	221	310	0.61

4.4.3. RUL prediction results and analysis

This section evaluates the RUL prediction performance of the proposed model using different input feature sets, with predictions starting at the 576th hour for FC1 and the 509th hour for FC2. Prognostics are conducted using 50 % of the sample dataset for training, and the real and predicted RULs under different FT conditions are calculated using Eq. (10) and Eq. (11). The RUL results are presented in Table 5 and Fig. 19 for FC1, and Table 6 and Fig. 20 for FC2.

From observing Table 5 and Fig. 19, it is clear that the optimal-input model provides better predictive accuracy, obtaining a final score of 0.94, which is 54.10 % higher than other input models with a final score of 0.61. Notably, the optimal-input model precisely matches real observations at the 4 % FT condition. Similarly, for FC2 (Table 6 and Fig. 20), the optimal-input model achieves a final score of 0.97, the

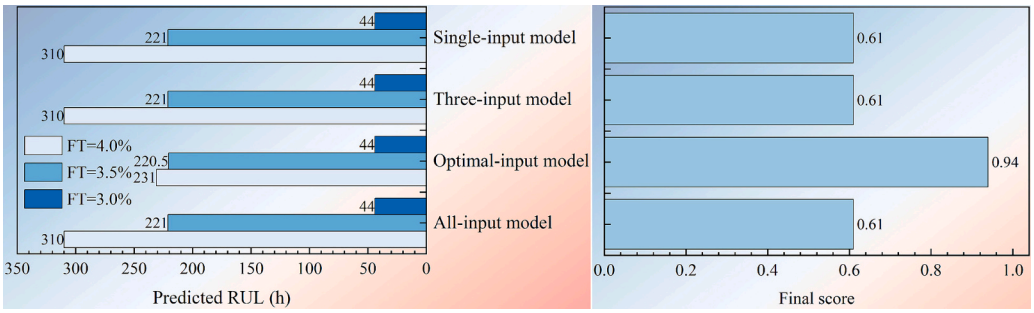


Fig. 19. RUL prediction results of four models for FC1.

Table 6
Comparison of RUL between real observations and prediction results obtained from models with different input feature sets, along with the corresponding final scores for FC2.

Models	RUL results (h)			Final score (*FAscore*)
	FT = 4.0 %	FT = 4.5 %	FT = 5.0 %	
Real observations	108	246.5	261.5	–
Single-input model	109	247	262.5	0.93
Three-input model	207.5	247.5	410.5	0.32
Optimal-input model	**108.5**	**247**	**261**	**0.97**
All-input model	207.5	247.5	261.5	0.65

closest to 1. Additionally, under an FT of 5.0 %, the predicted RUL of the optimal-input model is 0.5 h earlier than the actual failure threshold, which is beneficial for the maintenance of fuel cell systems. These results confirm that the predictive model with the optimal feature set predicts the RUL of fuel cells more accurately, further demonstrating the superiority of the proposed RF-TCN approach in predicting both FDT and RUL.

4.4.4. Comparison with other models

To further validate the effectiveness of the proposed RF-TCN approach, conventional deep learning models, including CNN, LSTM, and GRU, are employed to predict the FDT of PEMFC. In addition, predictions under different horizons (n steps ahead, with n = 5, 10, 15, 24) from similar work [19] are included for comparison. Moreover, the baseline TCN model [46], referred to as BTCN, is also implemented to assess the improvement of the improved TCN model. It is worth mentioning that the optimal feature set selected via the RF-RFE method is applied consistently across all models. The visual comparisons are depicted in Fig. 21 and Fig. 22 for FC1 and FC2, respectively, while Fig. 23 presents the comparative results of different models. MAPE and RMSE are used as evaluation metrics to ensure an unbiased comparison of prediction performances, as done in [18,19,65].

As depicted in Fig. 21 and Fig. 22, predictions begin at the 579th hour for FC1 and the 612th hour for FC2. The proposed RF-TCN approach best matches the real voltage degradation trends, followed by the BTCN model, which generally provides the second-best predictions. In contrast, the CNN model exhibits significant fluctuations, while the LSTM and GRU models observe substantial deviations from the actual degradation curves. Furthermore, compared to the results in [19] (Fig. 12 for FC1 and Fig. 13 for FC2 in [19]), which show significant irregular degradation trends and "late" predictions, the proposed approach more accurately captures the real degradation behaviour and provides "early" degradation predictions for the FC2 task, indicating that the proposed approach is suitable for handling long-time data, even with fluctuation in the raw data, and can identify early signs of degradation of fuel cells.

Accurate and "early" predictions are particularly valuable for control-oriented ageing prediction problems, as they provide operators with sufficient lead time to optimize system performance. While the accuracy of the CNN/LSTM/GRU models could be improved with attention mechanisms [18,19], such enhancements would increase the model complexity [18,23]. In contrast, the proposed approach achieves superior prediction accuracy without requiring additional complexity, making it a more practical and efficient solution.

Fig. 23 shows that the prediction performance reduces as the prediction horizon extends. This results from the models' inability to capture both local and global features from the input sequences [23]. Despite this, the proposed RF-TCN approach achieves higher prediction accuracy than all other models, including the attention-based dilated CNN developed in Ref. [19]. Moreover, compared to the RMSE reported in [66] for 10 steps ahead prediction horizon (0.00730 for FC1 and 0.00810 for FC2), the proposed RF-TCN approach obtains RMSE values of 0.00627 for FC1 and 0.00765 for FC2, improving the accuracy by 16.43 % and 5.88 %, respectively. Overall, it can be further concluded that the proposed RF-TCN approach is more capable of handling PEMFC prognostic tasks and is more valuable in practice.

The performance improvement of the improved TCN model compared to the BTCN model is presented in Table 7. The predictive performance has improved when using the improved TCN model. For instance, in the case of FC1, the MAPE and RMSE of the improved TCN model decreased by 5.72–45.15 % and 4.92–20.11 %, respectively.

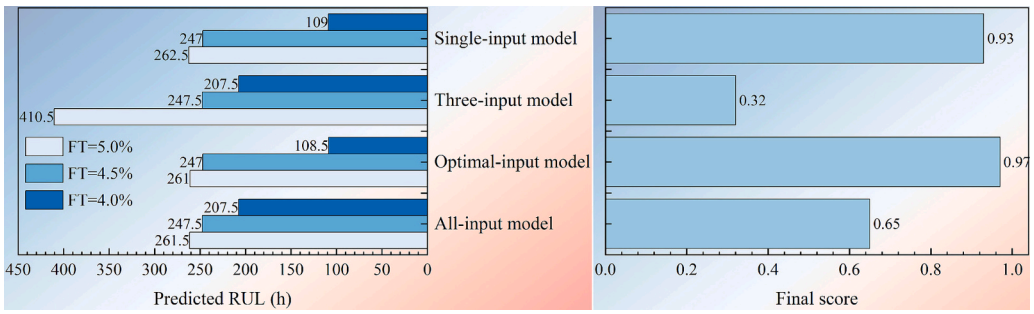


Fig. 20. RUL prediction results of four models for FC2.

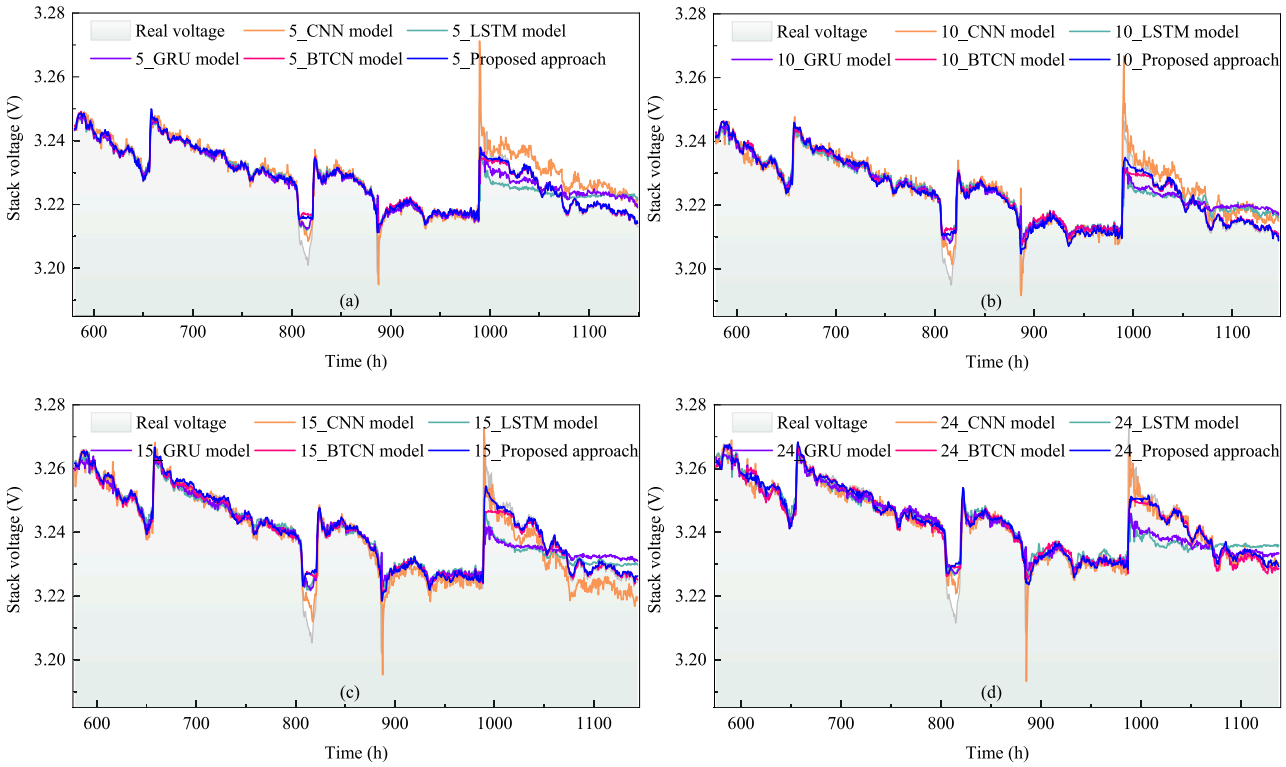


Fig. 21. Comparison of different models under various prediction horizons for FC1: (a) 5 steps ahead, (b) 10 steps ahead, (c) 15 steps ahead, and (d) 24 steps ahead.

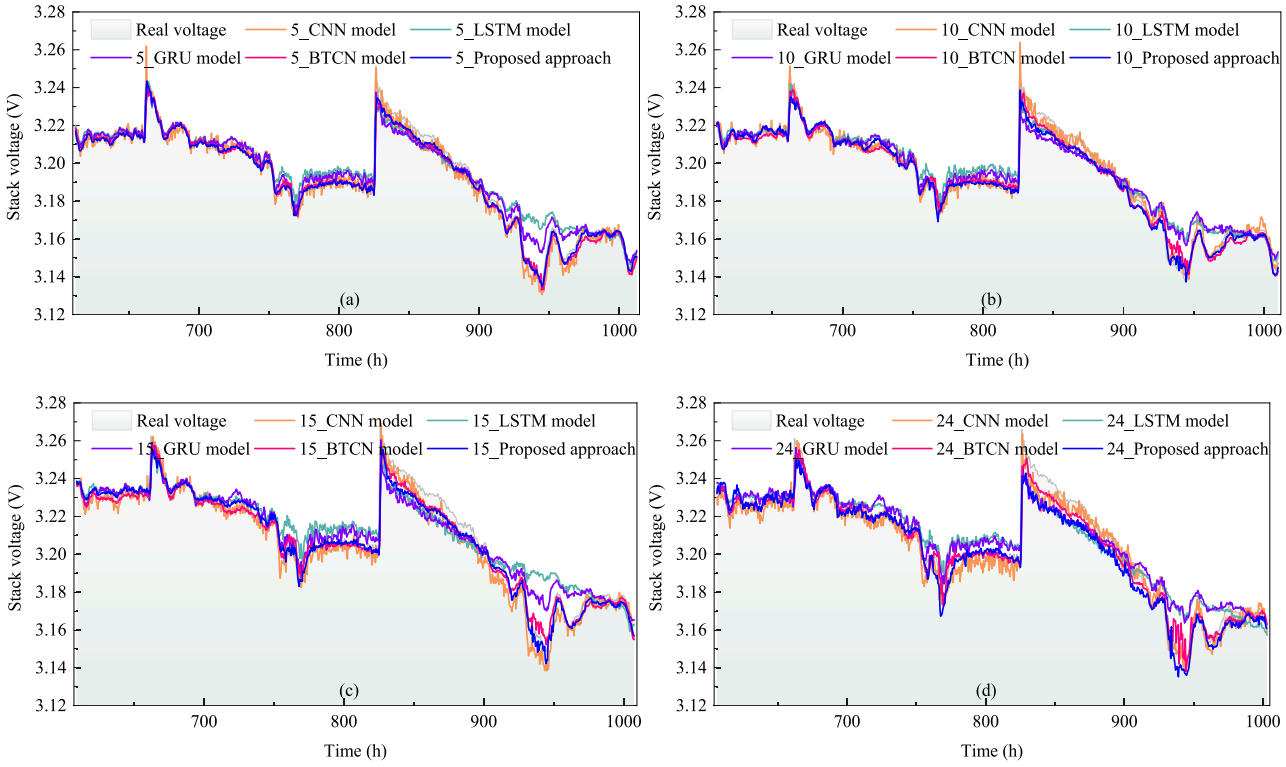


Fig. 22. Comparison of different models under various prediction horizons for FC2: (a) 5 steps ahead, (b) 10 steps ahead, (c) 15 steps ahead, and (d) 24 steps ahead.

These results confirm the effectiveness of the modifications introduced in this study and underscore the superiority of the improved TCN model for PEMFC prognostic tasks.

Table 8 compares the implementation time of the improved TCN model and the TCN model with repeating residual blocks described in [48,49], using identical parameters. The improved model achieves implementation time reductions of up to 37 % for FC1 and 52 % for FC2. While the improved model demonstrates notable computational efficiency in the tested scenarios, it may require longer computation time for large-scale datasets in practical applications. However, since PEMFC

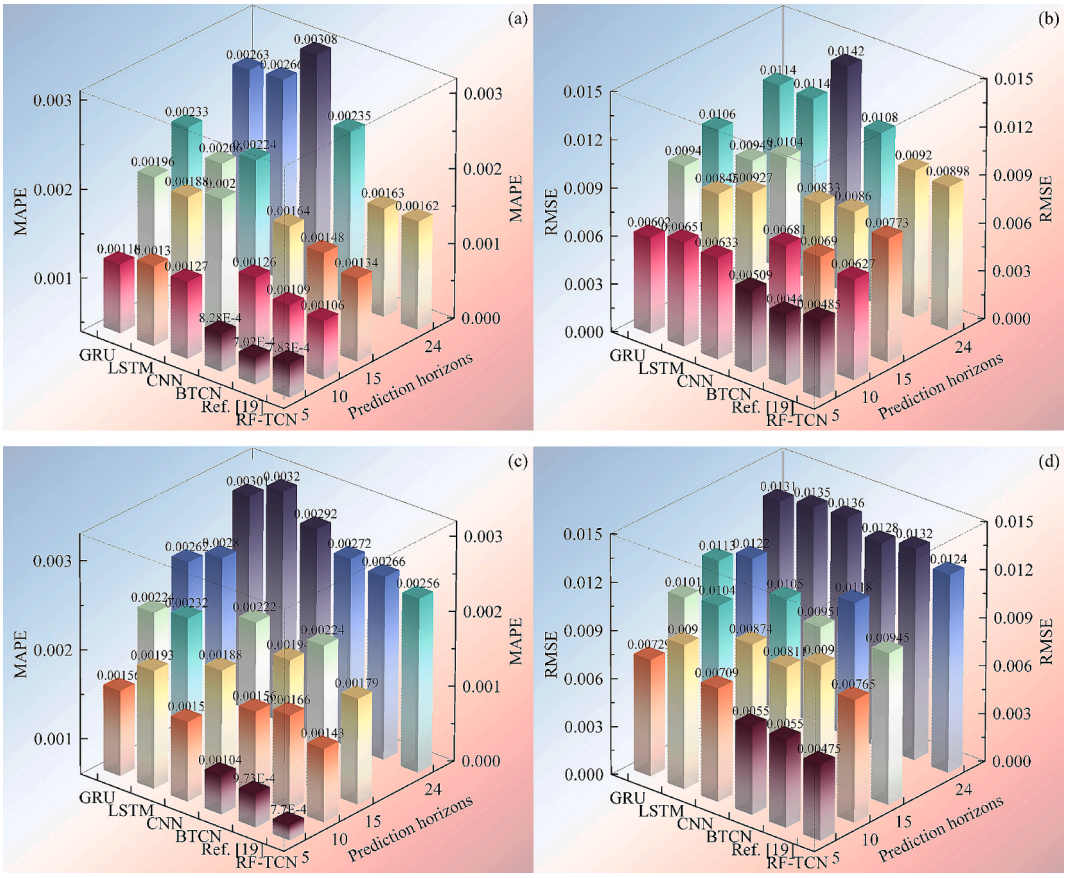


Fig. 23. Analysis of the effectiveness of the proposed RF-TCN approach for (a-b) FC1 and (c-d) FC2.

Table 7
Performance improvement of the improved TCN model compared to the BTCN model.

Datasets	PI_{index} (%)	Prediction horizons			
		5 steps ahead	10 steps ahead	15 steps ahead	24 steps ahead
FC1	PI_{MAPE}	5.72	18.78	22.50	45.15
	PI_{RMSE}	4.92	8.61	7.82	20.11
FC2	PI_{MAPE}	34.63	9.10	8.26	6.56
	PI_{RMSE}	15.85	5.94	0.63	3.37

Table 8
Implementation time comparison between the improved TCN model and the TCN model in [48,49].

Datasets	Models	Implementation time (s)			
		5 steps ahead	10 steps ahead	15 steps ahead	24 steps ahead
FC1	Improved TCN	78.61 **(↓37 %)**	79.31 **(↓35 %)**	78.20 **(↓37 %)**	78.23 **(↓37 %)**
	TCN [48,49]	107.38	106.80	107.25	107.45
FC2	Improved TCN	37.29 **(↓36 %)**	36.11 **(↓36 %)**	33.00 **(↓52 %)**	34.21 **(↓51 %)**
	TCN [48,49]	50.87	49.21	50.17	51.74

degradation and RUL predictions typically occur over extended timescales, rapid computation is not a primary requirement. Therefore, the improved TCN model offers a competitive edge in both accuracy and efficiency.

5. Conclusions

This paper introduces a novel multivariable prognostic approach for PEMFCs, termed RF-TCN, which combines random forest and temporal convolutional networks. The proposed approach is validated using two durability test datasets and compared with existing literature results [19], the BTCN model [46], and three traditional deep learning models.

Firstly, a hybrid feature selection method integrating the advantages of RF and RFE algorithms is employed to automatically select optimal features. These features are then validated against existing mechanistic analysis results and subsequently fed into the improved TCN-based predictive model optimized using the PSO algorithm. The model performance is evaluated under different training lengths and input feature sets, along with its multi-step ahead prediction capability. Results show that the model maintains high prediction accuracy across different training lengths, demonstrating robust generalization. When adopting optimal input features, the model achieves higher accuracy and reduced uncertainty compared to single-input and all-input models, verifying the effectiveness of selected features. Furthermore, the proposed approach outperforms the attention-based dilated CNN in [19], the BTCN model [46], and CNN, LSTM, and GRU models in PEMFC degradation prediction problems for multi-step ahead forecasting. These results demonstrate that the proposed approach is capable of effectively handling PEMFC prognostic tasks.

Overall, the data-driven prognostic approach developed in this work can precisely predict the FDT and RUL of fuel cells, which can assist in achieving intelligent decision-making processes for fuel cell systems, including real-time optimization of fault-tolerant control and management strategies.

CRediT authorship contribution statement

Tian Zhang: Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Zhengmeng Hou:** Writing – review & editing, Supervision, Conceptualization. **Xiaoqin Li:** Resources. **Qianjun Chen:** Investigation. **Qichen Wang:** Data curation. **Christian Lüddeke:** Visualization. **Lin Wu:** Validation. **Xuning Wu:** Formal analysis. **Wei Sun:** Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work is supported by the European Union's "Horizon Europe programme"-LOC3G (Grant No. 101129729), the HaiJu Plan for High-Level Talent Introduction Project of Sichuan Province (No. 2024JDHJ0012), and the High-End Foreign Experts Program of the Yunnan Revitalization Talents Support Plan of Yunnan Province. The first author Mr. Zhang would like to thank the China Scholarship Council (No. 202208530003) for fellowship support.

Data availability

Data will be made available on request.

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